Analysis of Failure Mechanisms During the Long-Term Retention Operation in 3-D NAND Flash Memories

Shinkeun Kim, Kyunghwan Lee, Changbeom Woo, Yuchul Hwang, *Member, IEEE*, and Hyungcheol Shin, *Senior Member, IEEE*

***Abstract*— Our previous study successfully separated the failure mechanisms that cause charge loss during long-term retention operation of 2-D NAND flash memories using the limiting conditions established by prior literature and experiments. Based on these previous research, we investigated the failure mechanisms during long-term retention (~1000 h) of 3-D NAND flash memories. Each mechanism was separated from the total charge loss measurement data. Program and erase (P/E) cycles deteriorate tunnel oxide quality and increase the contribution rate (CR) of the failure mechanisms associated with tunnel oxide. Our analysis uncovered the dependence that specific parameters related to tunnel oxide have on P/E cycles. It was confirmed that the charge loss function describes the real-world behavior of charge loss mechanisms well. Considering the conditions of how lateral charge loss mechanism occurs by 3-D NAND flash memories' structural features, the constraints set in previous research was updated, and the validity of these constraints was confirmed by analysis using a TCAD simulation.**

***Index Terms*— 3-D NAND flash, cycling dependence, detrapping, failure mechanism, interface trap recovery, lateral migration (LM), long-term retention, stretched exponential function, trap-assisted tunneling (TAT).**

I. Introduction

WHILE the innovation to the 3-D vertical structure has been implemented to reduce the bits cost [1]–[5], the charge loss during the retention time was still a troublesome problem for the reliability of 3-D NAND flash memories [6]–[8]. In particular, the move from the 2-D planar structure to a 3-D vertical structure requires a change in the materials and fabrication process of the charge trap layer (CTL), which caused a new failure mechanism to appear in addition to the failure mechanisms that causes the charge loss in the 2-D planar structure. Various failure mechanisms can cause the charge loss that occurs during the retention operation. It is difficult to accurately analyze the tendencies of charge loss, because these mechanisms were mixed to induce charge loss simultaneously [9], [10]. Our previous work used a charge loss function using the stretched exponential function to solve these difficulties and separate these failure mechanisms from each other [11]–[15]. The limiting conditions of each failure mechanism have since been established through several studies and various experiments. These coexisting mechanisms can now be separated using the charge loss function under the newly established limiting conditions.

In studies of planar structure, we reported that detrapping [10], [17]–[19], trap-assisted tunneling (TAT) [10], [20]–[23], interface trap (N_{it}) recovery [10], [24]–[26] mechanisms generally cause charge loss during long-term retention operations. Vertical structures cause charge migration in the lateral direction (LM) during retention operations because the target cell shares its CTL with adjacent cells [28]–[31]. Therefore, in 3-D NAND flash memories, there are four dominant failure mechanisms: three mechanisms have been reported in studies of planar structure memory, and one new mechanism in unique to vertical structure memory.

In this article, we analyze the charge loss characteristics during long-term retention operations of 3-D NAND flash memories by analyzing the four failure mechanisms in Section II. Analysis and separation of the failure mechanism were carried out under new limiting conditions of 3-D NAND flash memories using the same charge loss function exploited in previous research. This new limiting condition for 3-D NAND flash memories is presented by adding the LM parameter to the 2-D planar constraints reported in a previous article [16]. In Section III, the updated limiting conditions on each failure mechanism's parameters are verified using a TCAD simulation. We set up the simulation environment for the four mechanisms and compared the individual results from the charge loss function with the simulation results to confirm the new constraint's validity. In Section IV, the retention characteristics over the long term are analyzed according to the number of program and erase (P/E) cycles. The failure mechanisms are separated into various P/E cycles' need considering the tunnel oxide degradation caused by these P/E cycles. By analyzing each mechanism's parameter trends, we investigate long-term retention characteristics through the degradation caused by P/E cycles.

Manuscript received August 10, 2020; revised September 14, 2020; accepted September 28, 2020. Date of publication October 19, 2020; date of current version November 24, 2020. The review of this article was arranged by Editor P.-Y. Du. *(Corresponding author: Shinkeun Kim.)*

Shinkeun Kim, Changbeom Woo, and Hyungcheol Shin are with the Inter-University Semiconductor Research Center, Seoul National University, Seoul 08826, South Korea, and also with the School of Electrical Engineering and Computer Science, Seoul National University, Seoul 08826, South Korea (e-mail: skeunk@snu.ac.kr).

Kyunghwan Lee is with the Semiconductor Research and Development Center, Flash TD Team, Samsung Electronics, Hwaseong 18448, South Korea.

Yuchul Hwang is with the Memory Division, Quality Assurance Team, Samsung Electronics, Hwaseong 18448, South Korea.

Color versions of one or more of the figures in this article are available online at https://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TED.2020.3028349

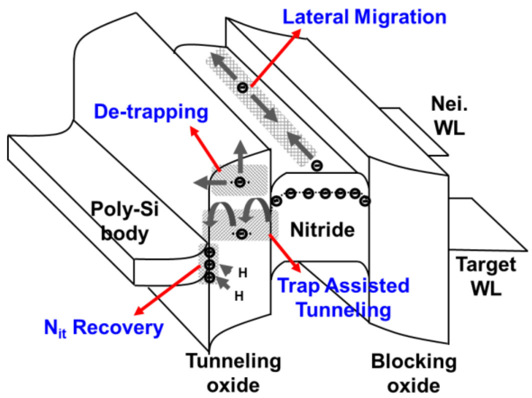


Fig. 1. Schematic energy band diagram of 3-D NAND flash memory in programed target cells during long-term retention operation.

II. Separation of Failure Mechanism in 3-D NAND Flash Memory

In this section, we used the charge loss function used in previous studies [11]–[16] to separate failure mechanisms from main chip measurements at a certain probability level $(P) = 0.1$ for 3-D NAND flash memories. ΔV_{th} caused by charge loss can be expressed as the sum of each of the four mechanisms' functions. In this study, in addition to the three mechanisms reported for 2-D planar NAND, the long-term retention characteristics are analyzed based on the separated results that take into account the charge loss by LM. We also update the limiting conditions for the 3-D vertical structure.

Fig. 1 schematically describes the charge loss components of a programmed target cell during long-term retention operation. In this work, detrapping, TAT, N_{it} recovery, and LM are considered significant failure mechanisms that cause charge loss during long-term retention operations. $\Delta V_{th,total}$ during long-term retention operation is the sum of each charge loss function and can be separated as in [11]–[16]

$$\Delta V_{th,Total} = \sum_{k=1}^{4} \Delta V_{th,mechanism(k)} \times \left(1 - \exp\left(-\left(\frac{t_R}{\tau_{mechanism(k)}}\right)^{\beta_{mechanism(k)}}\right)\right) \quad (1)$$

where t_R, $\Delta V_{th,mechanism(k)}$, $\tau_{mechanism(k)}$, and $\beta_{mechanism(k)}$ are the retention time, final value of $\Delta V_{th}$for each mechanism, time-constant, and shape parameter of charge loss function for each mechanism, respectively. The parameters of each mechanism's function were determined by following the limiting conditions shown in Table I. In this article, the limiting conditions to separate the failure mechanisms were established by referring to the parameter extraction method used for 2-D planar devices [16] from our group. In a 3-D vertical structure, the amount of charge loss that occurs during retention operation was reported to be greater than that of a 2-D planar structure [11]–[16]. 2-D and 3-D NAND are known to differ significantly in their fabrication process, and their resulting characteristics are also known to differ significantly. However, from the perspective of their retention characteristics, most phenomena are founded to impact on short-term retention operation remarkably [8]. They are considered to have relatively little effect on long-term retention operation. Thus, we think that the only difference in the long-term retention characteristics between 2-D and 3-D would come from a charge loss mechanism in the horizontal direction. It was expected that detrapping and N_{it} recovery would contribute less to the total charge loss than TAT and LM. Considering each mechanism's charge source, the amount of electrons stored in the CTL is much more considerable than the other trap sites. Therefore, the final V_{th} of each mechanism can be determined, as shown in Table I, which considers the expected more enormous contributions of LM and TAT and the results of research on 2-D NAND [16]. τ_{LM} is expected to be larger than that from detrapping and smaller than that from TAT. Although LM is caused by the migration of electrons within the CTL, it is not affected by barriers in the material, so it seems to occur relatively quickly compared with TAT, which needs traps to cross the barrier. Also, τ_{LM} seems more considerable than that from detrapping because there are many more electrons trapped in the CTL than electrons trapped in the tunnel oxide. Fig. 2 shows the 3.0 k P/E cycled measurement data at $P = 0.1$ and the separated results corresponding to different temperatures (T). The symbol represents the results measured at the main chip, and the solid color lines over time represent the value of $\Delta V_{th,total}$ by each mechanism. The black solid line shows the total charge loss's best fit as the sum of each mechanism's functions. As temperature increases, $\Delta V_{th,total}$ caused by the charge loss increases, and the contribution rates (CRs) of each failure mechanism change with temperature and time.

TABLE I
Summary of Limiting Condition for the Separation of Failure Mechanisms

Limiting Condition For Separation	
▪ $\Delta V_{th(Nit)} < \Delta V_{th(De\text{-}trapping)} < \Delta V_{th(TAT)} < \Delta V_{th(LM)}$	
▪ $\Delta V_{th(LT)} \leq \Delta V_{th(HT)}$	
▪ $\beta_{TAT} < \beta_{De\text{-}trapping} < 1$	▪ $\ln \tau_{(HT)} \leq \ln \tau_{(LT)}$
▪ $\beta_{TAT} < \beta_{Nit} < 1$	▪ $\tau_{Nit@125°C} < 10\text{hrs}$
▪ $\beta_{LM} < \beta_{TAT} < 1$	▪ $\tau_{Nit} < \tau_{De\text{-}trapping} < \tau_{LM} < \tau_{TAT}$
▪ $\beta_{De\text{-}trapping(125°C)} = \beta_{De\text{-}trapping(100°C)} = \beta_{De\text{-}trapping(85°C)} \cdots$	
▪ $\beta_{Nit(125°C)} = \beta_{Nit(100°C)} = \beta_{Nit(85°C)} \cdots$	

III. Validation of Parameter Constraints

In this section, the updated limiting conditions from Section II have been validated through the Synopsys Sentaurus TCAD simulation tool. This verification was performed by comparing the TCAD simulation results with the charge loss function for a mechanism separation using limiting conditions. The device structure and parameters are shown in Fig. 3. The spacer length (L_s) and the gate length of the target cell (L_{WL}) used one of the values within the range shown in Table II. Table II shows the model parameters used in each mechanism and the parameters used in common. According to the mechanisms, the trap energy level (E_T), trap density (N_T), and capture cross section (σ) were changed to compare the separation results by the charge loss model with TCAD simulation.

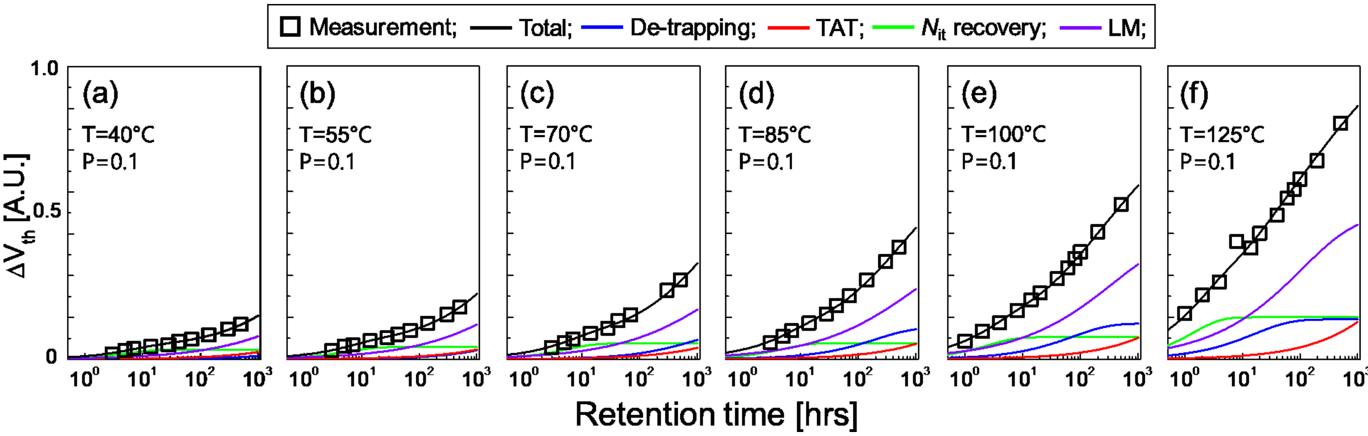
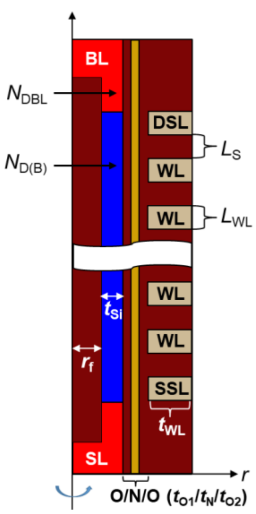


Fig. 2. Separated results of failure mechanisms from 3 k P/E cycled measurement data ($P = 0.1$) for a PV7 state of a 3-D NAND main chip, according to various temperature (T) values. (a) $T = 40$ °C. (b) $T = 55$ °C. (c) $T = 70$ °C. (d) $T = 85$ °C. (e) $T = 100$ °C. (f) $T = 125$ °C.



Device parameter	
L_{WL}	30 ~ 50 nm
L_S	30 ~ 50 nm
t_{si}	10 nm
r_f	25 nm
O/N/O	5 / 5 / 6 nm
$N_{D(B)}$	10^{15} cm^{-3}
$N_{DBL} = N_{DSL}$	10^{19} cm^{-3}

Fig. 3. Schematic cross section of simulation structure and device parameters range. The device parameters are as follows: L_{WL} is the target cell gate length range, L_s is the spacer length range, t_{si} is the channel thickness, r_f is the filler-oxide radius. O/N/O indicate the thickness of tunnel oxide, CTL, and blocking oxide, respectively, and $N_{D(B)}$, N_{DBL}, and N_{DSL} are the donor doping concentration of the bit line, source line, and channel, respectively.

TABLE II
MODEL PARAMETERS USED IN THIS SIMULATION

Mechanism	parameter	Value
Common	N_{T_CTL}	1x10^{19} cm^{-3}
	E_{T_CTL}	1.8 eV
	σ_{CTL}	10^{-15} cm^2
De-trapping	$N_{T\_tunnel.Ox}$	2.4x10^{18} cm^{-3}
	$E_{T\_tunnel.Ox}$	1.282 eV
	$\sigma_{tunnel.Ox}$	10^{-14} cm^2
TAT	$E_{T\_tunnel.Ox}$	1.24 eV
LM	E_{T_CTL}	0.78eV
	σ_{CTL}	10^{-13} cm^2
N_{it} Recovery	N_0	5x10^{12} cm^{-2}
	k_{F0}	70 cm^3/s
	k_{R0}	9.9x10^{-6} cm^3/s
	E_{ADH}	0.2 eV

A. Detrapping Simulation

The detrapping mechanism describes electrons being inadvertently trapped in tunnel oxide during the program. Fig. 4(a) shows electrons detrapping from tunnel oxide over time in the 3-D NAND flash structure. The trap density (N_t) was set by referring to the value of $\Delta V_{th(Detrapping)}$ extracted in Section II. This analysis was performed by trapping electrons through program operations in the tunnel oxide. The electrons trapped in the tunnel oxide were detrapped based on the Poole–Frenkel (PF) and nonlocal tunneling model included in the TCAD simulation. As shown in Fig. 4(b), the simulation results are well matched at all temperatures with the charge loss function results for detrapping in Section II.

B. TAT Simulation

The electrons trapped in the CTL could more easily tunnel toward the channel with the help of a single trap in the tunnel oxide. As shown in Fig. 4(c), this mechanism was implemented through TCAD simulation. In this case, the tunneling occurs only through a single trap in the tunnel oxide; all other components that contribute to tunneling were removed. The number of electrons trapped in the CTL was determined regarding $\Delta V_{th(TAT)}$, like in the case of detrapping. This mechanism was simulated using the nonlocal tunneling model to describe the tunneling from the CTL and a single trap. Fig. 4(d) shows that the simulation results and the charge loss function results well matched at all temperatures.

C. N_{it} Recovery Simulation

The hydrogen atoms recovered N_{it} generated by the P/E cycle during long-term retention operation. This phenomenon is considered a failure mechanism in the same way as the other mechanisms because it reduces $\Delta V_{th,total}$. The multistate configuration (MSC)-hydrogen transport degradation model built within TCAD was used for simulation of this mechanism. The values of the initial Si-H bond density (N_0), forward and reverse reaction rate constants (k_{F0} and k_{R0}), and activation energy in diffusion (E_{ADH}) parameters were used to compare the separation results for N_{it} recovery mechanism, as shown in Table II. Fig. 4(e) shows decreased hydrogen atoms' density due to their combination with interface traps during retention operation. In the same way as previous results, Fig. 4(f) shows that the $\Delta V_{th(Nitrecovery)}$ trend by simulation well-matched with the charge loss function results.

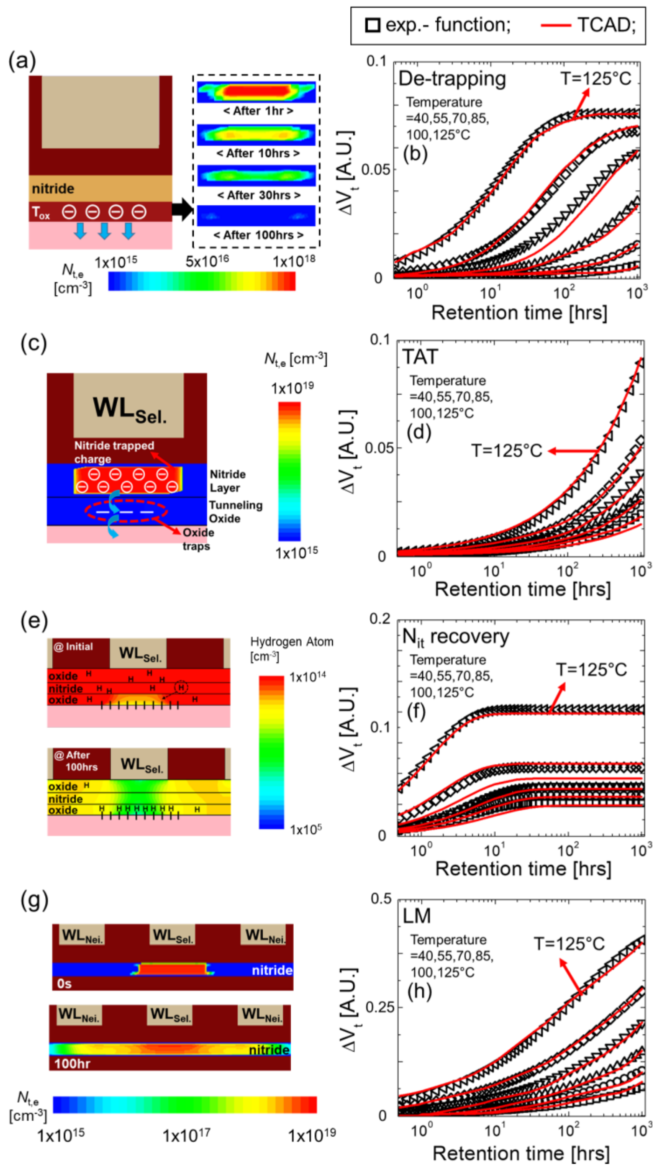


Fig. 4. TCAD simulation environment for the four failure mechanisms and a comparison of the TCAD simulation results with the charge loss function results at various temperatures. (a) Detrapping simulation. (b) Comparison of two detrapping results. (c) TAT simulation. (d) Comparison of two TAT results. (e) N_{it} recovery simulation. (f) Comparison of two N_{it} recovery results. (g) LM simulation. (h) Comparison of two LM results.

D. LM Simulation

In this article, only the lateral direction was considered for the direction of charge migration in the CTL. Previous research in our groups has reported vertical charge redistribution (VR) in the CTL [8]. However, since VR has already occurred and becomes saturated by the short-term operation, this phenomenon does not affect its long-term characteristics. The simulation of this mechanism was performed and did not include any vertical component. This simulation took into account the movement of electrons inside the CTL using the PF model. Fig. 4(g) shows that the electrons trapped in the CTL move in a lateral direction during the retention operation. $\Delta V_{th(LM)}$ from the TCAD simulation is in good agreement with the charge loss function results, as shown in Fig. 4(h).

Each simulation focused on only one mechanism at a time; assuming each mechanism occurs independently, we can confirm that the simulation results well-matched with the charge loss function results for each mechanism.

IV. Analysis of P/E Cycle Dependence on Retention Characteristics

Based on the limiting conditions validated by TCAD simulation in Section III, each failure mechanism was separated from the measurement data for various P/E cycle times in addition to the 3.0 k results previously discussed in Section I. The tendencies of the parameters in the charge loss function for each mechanism, according to temperatures ($T = 40$ °C, 55 °C, 70 °C, 85 °C, 100 °C, and 125 °C) and several cycles (0.5 k, 1.0 k, 3.0 k, and 5.0 k), were analyzed to investigate the effect of the P/E cycles on the retention characteristics of 3-D NAND. As the P/E cycle stress is applied, the trap is generated on the tunnel oxide and tunnel oxide/channel interface, and as the number of P/E cycles increases, more traps are generated. The failure mechanisms associated with these increased traps contribute more to the charge loss during the retention operation. In particular, we focused on the tunnel oxide-related mechanisms that were expected to be significantly affected by the P/E cycle.

Fig. 5 shows the separation results expressed as the sum of the functions for each mechanism using the parameters determined according to the limiting conditions from the results measured from the 3-D NAND main chip ($P = 0.1$) for two temperatures (at 40 °C and 125 °C) over 0.5, 1.0, and 5.0 k P/E cycled. As the number of P/E cycles increases, we can see that $\Delta V_{th,total}$ caused by the charge loss increases, and the change caused by increasing the cycle time is more remarkable at high temperatures (HTs). As with the 3.0 k results in Fig. 2, as the temperature increased, the value of $\Delta V_{th,total}$ increased, and the CR of each mechanism changed. Besides, the number of P/E cycles significantly affects the composition of each mechanism. When comparing Fig. 5(d)–(f), it is clear that the CR of each mechanism changes depending on the number of P/E cycles. Regardless of the number of P/E cycles, LM contributes the most to the charge loss during the long-term retention. However, CR_{LM} tends to decrease compared with other mechanisms. $\Delta V_{th,total}$ increases despite little change in LM as the number of P/E cycles increases because the failure mechanisms affected by tunnel oxide degradation increase the amount contributing to the charge loss. Among these mechanisms, detrapping and N_{it} recovery show the most remarkable changes, contributing more and more quickly to the charge loss as the number of P/E cycles increases. As the number of P/E cycles increases, the charge loss caused by TAT is also increasing. By analyzing the charge loss function parameters used for mechanism separation according to the number of P/E cycles, we can investigate the characteristics of the failure mechanisms that were expected to be affected by tunnel oxide degradation.

Fig. 6 shows the characteristics of $\beta_{mechanism(k)}$ according to temperature and P/E cycles. Fig. 6(a) and (c) shows that $\beta_{mechanism(k)}$ has a little dependence on temperature, as shown in previous studies [11]–[16] and Table I.

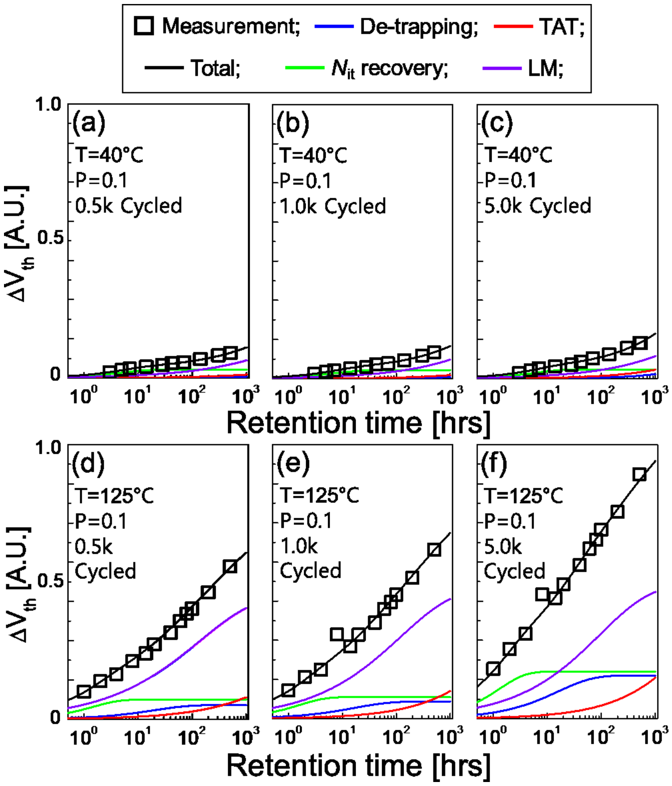


Fig. 5. Separated results of each failure mechanism according to P/E cycle number from data measured (P = 0.1) from PV7 of the 3-D NAND main chip with various T values. (a) 0.5 k P/E cycled in T = 40 °C. (b) 1.0 k P/E cycled in T = 40 °C. (c) 5.0 k P/E cycled in T = 40 °C. (d) 0.5 k P/E cycled in T = 125 °C. (e) 1.0 k P/E cycled in T = 125 °C. (f) 5.0 k P/E cycled in T = 125 °C.

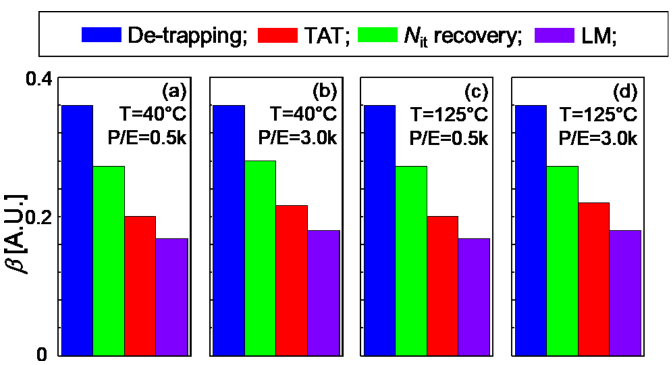


Fig. 6. Value of $\beta_{\mathrm{mechanism}(k)}$ parameter for each mechanism. (a) 0.5 k P/E cycled in T = 40 °C. (b) 3.0 k P/E cycled in T = 40 °C. (c) 0.5 k P/E cycled in T = 125 °C. (d) 3.0 k P/E cycled in T = 125 °C.

Also, Fig. 6(c) and (d) shows that there is also minimal dependence on P/E cycles.

In conclusion, $\beta_{\mathrm{mechanism}(k)}$ shows the same value at the number of P/E cycles according to the increasing temperature. $\beta_{\mathrm{mechanism}(k)}$ in the charge loss function is a parameter that is related to the varying contribution of each mechanism, so there is no change with temperature [16]. As with temperature, the number of P/E cycles does not affect each mechanism's variation. For long-term retention operation, it can be concluded that there was little impact on the variety of mechanisms caused by either temperature or number of P/E cycles. In the case of the floating gate (FG), the electrons

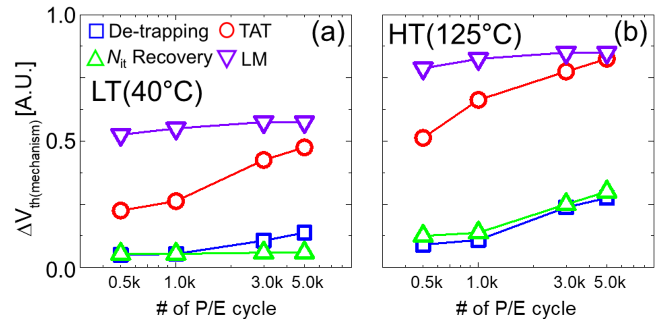


Fig. 7. Final Vth (mechanism) parameter tendency of each mechanism with increasing the number of P/E cycles (from 0.5 to 5.0 k). (a) T = 40 °C. (b) T = 125 °C.

stored in the FG spread quickly within the FG, and the potential is the same. Whereas in the case of the CTL in 3-D NAND, the electrons clustered in the CTL center spread toward spacers. LM is carried out until the electric field caused by the difference between the center and edge of the CTL is weak. On the other hand, TAT continues to be caused by a vertical field caused by charges stored in CTL, even if the lateral electric field is resolved. The reason why β_{LM} is a little smaller than β_{TAT} because the moving path of LM is longer than the vertical direction path of TAT.

Fig. 7(a) shows a tendency for the final $V_{\mathrm{th(mechanism)}}$ value of each mechanism with increasing the number of P/E cycles at a relatively low temperature (LT) of 40 °C. Fig. 7(b) shows the case at an HT of 125 °C. It seems that increasing the number of P/E cycles generates traps in the tunnel oxide bulk, which increases the source of the detrapping mechanism. However, this increased source of detrapping is relatively less affected by temperature [31]. Therefore, the $\Delta V_{\mathrm{th(detrapping)}}$ increases with the number of P/E cycles, while there is little change in temperature. As in the detrapping case, TAT was also affected by the degradation of the tunnel oxide. In particular, the number of traps that could assist tunneling increases. $\Delta V_{\mathrm{th(TAT)}}$ also increases with the number of P/E cycles. In addition, $\Delta V_{\mathrm{th(TAT)}}$ increases with temperature because the charge loss source increases with temperature, unlike detrapping [31]. In the case of N_{it} recovery, the final V_{th} value was not significantly affected by the number of P/E cycles at LT, but the effect of P/E cycling was increased at the higher temperature. Although the tunnel oxide/channel interface is degraded by the P/E cycle to increase N_{it}, this trend occurs because the N_{it} recovery is greatly affected by temperature during the retention operation. Therefore, N_{it} recovery is relatively less influential in the low cycled case and contributes to high cycled cases. The three mechanisms, not including LM, were directly affected by the degradation of tunnel oxide, whereas LM appears to be less affected by the number of P/E cycles because LM only occurs inside the CTL. Therefore, $\Delta V_{\mathrm{th(LM)}}$ has little change in the number of P/E cycles in Fig. 7. As the temperature increases, the CTLs trapped charges can move more easily, so $\Delta V_{\mathrm{th(LM)}}$ increases with temperatures.

Fig. 8(a) and (b) shows the value $\tau_{\mathrm{mechanism}(k)}$ at LT and HT, respectively. $\tau_{\mathrm{mechanism}(k)}$ decreases as the temperature increases by the limiting conditions introduced in Table I. Also, $\tau_{\mathrm{mechanism}(k)}$ is affected by the number of P/E cycles,

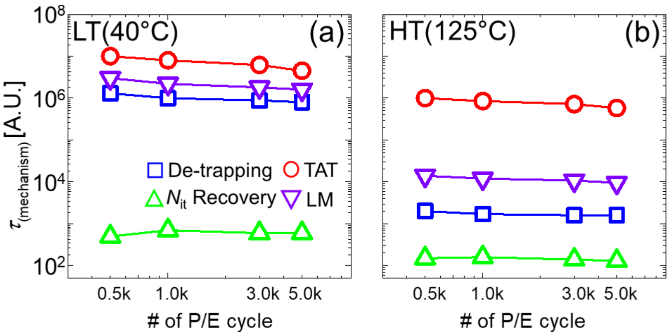


Fig. 8. τ(mechanism) parameter tendency of each mechanism with increasing the number of P/E cycles (from 0.5 to 5.0 k). (a) $T = 40$ °C. (b) $T = 125$ °C.

decreasing overall, but significantly less than the effect of temperature. Fig. 8 allows us to infer the activation energy (E_a) of each mechanism. E_a of detrapping is expected to be the largest because of the greatest $\tau_{\mathrm{mechanism}(k)}$ change according to temperature. Both TAT and LM are the trap-to-trap movement mechanisms, but compared with E_a, E_a of LM is much larger than that of TAT. That is because TAT is more dependent on tunneling due to the large energy barrier on the tunnel oxide, resulting in a greater vertical electric field, which results in lower temperature dependence and larger $\tau_{\mathrm{mechanism}(k)}$. After all, the trap density is much lower than the CTL layer. On the other hand, in the case of LM, the distance between traps is shorter because of the abundance of traps in the CTL, and thermal dependence is much higher because it is a hopping mechanism that is transferred to the next trap site by horizontal electric field after receiving the thermal energy. Considering the results in Fig. 8(a) and (b), the detrapping was most affected by temperature compared with the other mechanisms [16]–[19]. That is because detrapping is influenced by thermal emissions and direct tunneling due to thin tunnel oxide at the same time, but the probability that detrapping is still determined by temperature. The number of trap sites is thought to increase according to tunnel oxide degradation in the TAT case. This increase in the trap site can be thought of as increasing the path through which TAT occurs. Considering these characteristics, it follows that the dependence of τ_{TAT} on the number of P/E cycles is more extensive than other mechanisms. On the other hand, LM has relatively little τ_{LM} change by the P/E cycle, because the CTL already has many trap sites since the initial state. Similarly, in the case of N_{it} recovery, it can be seen that there is little dependence on the number of P/E cycles for $\tau_{N\mathrm{it\_recovery}}$. As discussed in Section I, N_{it} recovery is a high-speed mechanism, so even as the number of P/E cycles increases, the absolute amount that contributes to the total charge loss may increase, but when this occurs is not earlier.

V. Conclusion

In this article, the failure mechanisms were separated from measurement data to investigate the characteristics of the long-term retention of 3-D NAND flash memory. The limiting conditions were newly constructed to account for the lateral migration mechanism caused by a 2-D planar to 3-D vertical structure. TCAD simulation was used to verify that the revised limiting conditions well reflected each failure mechanism's physical characteristics. The failure mechanisms contributing to charge loss during long-term retention operations were successfully separated at various temperatures through the validated constraints. Also, the separated results according to the number of P/E cycles were used to analyze how the degradation caused by P/E repetition during long-term retention could affect each failure mechanism. The three mechanisms related to tunnel oxide were found to give an increased contribution to charge loss and advance the time these contributions occurred by analyzing the separating process parameters.

References

[1] E.-S. Choi and S.-K. Park, "Device considerations for high density and highly reliable 3D NAND flash cell in near future," in *IEDM Tech. Dig.*, Dec. 2012, p. 9, doi: 10.1109/IEDM.2012.6479011.

[2] M. Ishiduki *et al.*, "Optimal device structure for pipe-shaped BiCS flash memory for ultra high density storage device with excellent performance and reliability," in *IEDM Tech. Dig.*, Dec. 2009, p. 27, doi: 10.1109/IEDM.2009.5424261.

[3] J. Jang *et al.*, "Vertical cell array using TCAT (Terabit Cell Array Transistor) technology for ultra high density NAND flash memory," in *IEEE Symp. VLSI Tech. Dig.*, Jun. 2009, pp. 192–193.

[4] H. Tanaka *et al.*, "Bit cost scalable technology with punch and plug process for ultra high density flash memory," in *Proc. IEEE Symp. VLSI Technol.*, Jun. 2007, pp. 14–15, doi: 10.1109/VLSIT.2007.4339708.

[5] Y. Fukuzumi *et al.*, "Optimal integration and characteristics of vertical array devices for ultra-high density, bit-cost scalable flash memory," in *IEDM Tech. Dig.*, Dec. 2007, pp. 449–452, doi: 10.1109/IEDM.2007.4418970.

[6] B. Choi *et al.*, "Comprehensive evaluation of early retention (fast charge loss within a few seconds) characteristics in tube-type 3-D NAND flash memory," in *Proc. IEEE Symp. VLSI Technol.*, Jun. 2016, pp. 1–2, doi: 10.1109/VLSIT.2016.7573385.

[7] H.-J. Kang *et al.*, "Comprehensive analysis of retention characteristics in 3-D NAND flash memory cells with tube-type poly-Si channel structure," in *Proc. Symp. VLSI Technol. (VLSI Technol.)*, Jun. 2015, pp. T182–T183, doi: 10.1109/VLSIT.2015.7223670.

[8] C. Woo *et al.*, "Modeling of charge loss mechanisms during the short term retention operation in 3-D NAND flash memories," in *Proc. Symp. VLSI Technol.*, Jun. 2019, pp. T214–T215, doi: 10.23919/VLSIT.2019.8776579.

[9] B. Govoreanu and J. van Houdt, "On the roll-off of the activation energy plot in high-temperature flash memory retention tests and its impact on the reliability assessment," *IEEE Electron Device Lett.*, vol. 29, no. 2, pp. 177–179, Feb. 2008, doi: 10.1109/LED.2007.914089.

[10] *Failure Mechanisms and Models for Semiconductor Devices*, JEDEC Standard TEP122C, Mar. 2006.

[11] K. Lee, D. Kang, S. Seo, D.-H. Li, J. Kim, and H. Shin, "Analysis of failure mechanisms and extraction of activation energies (E_a) in 21-nm nand flash cells," *IEEE Electron Device Lett.*, vol. 34, no. 1, pp. 48–50, Jan. 2013, doi: 10.1109/LED.2012.2222013.

[12] K. Lee, D. Kang, H. Shin, S. Kwon, S. Kim, and Y. Hwang, "Analysis of failure mechanisms in erased state of sub 20-nm NAND Flash memory," in *Proc. 44th Eur. Solid State Device Res. Conf. (ESSDERC)*, Sep. 2014, pp. 58–61, doi: 10.1109/ESSDERC.2014.6948757.

[13] D. Kang *et al.*, "Generation dependence of retention characteristics in extremely scaled NAND flash memory," *IEEE Electron Device Lett.*, vol. 34, no. 9, pp. 1139–1141, Sep. 2013, doi: 10.1109/LED.2013.2271351.

[14] D. Kang *et al.*, "Probability level dependence of failure mechanisms in sub-20 nm NAND flash memory," *IEEE Electron Device Lett.*, vol. 35, no. 3, pp. 348–350, Mar. 2014, doi: 10.1109/LED.2014.2301164.

[15] D. Kang, K. Lee, S. Kwon, S. Kim, Y. Hwang, and H. Shin, "Analysis of read disturbance mechanism in retention of sub-20 nm NAND flash memory," *Jpn. J. Appl. Phys.*, vol. 54, no. 4S, Apr. 2015, Art. no. 04DD03, doi: 10.7567/JJAP.54.0D4DD03.

[16] K. Lee, M. Kang, Y. Hwang, and H. Shin, "Accurate lifetime estimation of sub-20-nm NAND flash memory," *IEEE Trans. Electron Devices*, vol. 63, no. 2, pp. 659–667, Feb. 2016, doi: 10.1109/TED.2015.2509004.

[17] C. M. Compagnoni *et al.*, "Investigation of the threshold voltage instability after distributed cycling in nanoscale NAND flash memory arrays," in *Proc. IEEE Int. Rel. Phys. Symp.*, May 2010, pp. 604–610, doi: 10.1109/IRPS.2010.5488762.

[18] Z. Xia *et al.*, "Comprehensive modeling of NAND flash memory reliability: Endurance and data retention," in *Proc. IEEE Int. Rel. Phys. Symp. (IRPS)*, Apr. 2012, p. MY-5, doi: 10.1109/IRPS.2012.6241922.

[19] G. Verma and N. Mielke, "Reliability performance of ETOX based flash memories," in *Proc. 26th Annu. Proc. Rel. Phys. Symp.*, Apr. 1988, pp. 158–166, doi: 10.1109/RELPHY.1988.23444.

[20] D. Ielmini, A. S. Spinelli, A. L. Lacaita, and A. Modelli, "A statistical model for SILC in flash memories," *IEEE Trans. Electron Devices*, vol. 49, no. 11, pp. 1955–1961, Nov. 2002, doi: 10.1109/TED.2002.804730.

[21] R. Degraeve *et al.*, "Analytical percolation model for predicting anomalous charge loss in flash memories," *IEEE Trans. Electron Devices*, vol. 51, no. 9, pp. 1392–1400, Sep. 2004, doi: 10.1109/TED.2004.833583.

[22] L. Larcher, "Statistical simulation of leakage currents in mos and flash memory devices with a new multiphonon trap-assisted tunneling model," *IEEE Trans. Electron Devices*, vol. 50, no. 5, pp. 1246–1253, May 2003, doi: 10.1109/TED.2003.813236.

[23] A. Padovani, L. Larcher, G. Bersuker, and P. Pavan, "Charge transport and degradation in HfO_2 and HfO_x dielectrics," *IEEE Electron Device Lett.*, vol. 34, no. 5, pp. 680–682, May 2013, doi: 10.1109/LED.2013.2251602.

[24] J.-D. Lee, J.-H. Choi, D. Park, and K. Kim, "Effects of interface trap generation and annihilation on the data retention characteristics of flash memory cells," *IEEE Trans. Device Mater. Rel.*, vol. 4, no. 1, pp. 110–117, Mar. 2004, doi: 10.1109/TDMR.2004.824360.

[25] C.-C. Lu, K.-S. Chang-Liao, F.-H. Tsai, C.-C. Li, and T.-K. Wang, "Detection of stress-induced interface trap generation on high-k gated nMOSFETs in real time by stress-and-sense charge pumping technique," *IEEE Trans. Electron Devices*, vol. 62, no. 5, pp. 1405–1410, May 2015, doi: 10.1109/TED.2015.2409477.

[26] S. Mahapatra *et al.*, "A comparative study of different physics-based NBTI models," *IEEE Trans. Electron Devices*, vol. 60, no. 3, pp. 901–916, Mar. 2013, doi: 10.1109/TED.2013.2238237.

[27] A. E. Islam, H. Kufluoglu, D. Varghese, S. Mahapatra, and M. A. Alam, "Recent issues in negative-bias temperature instability: Initial degradation, field dependence of interface trap generation, hole trapping effects, and relaxation," *IEEE Trans. Electron Devices*, vol. 54, no. 9, pp. 2143–2154, Sep. 2007, doi: 10.1109/TED.2007.902883.

[28] C. Kang *et al.*, "Effects of lateral charge spreading on the reliability of TANOS (TaN/AlO/SiN/Oxide/Si) NAND flash memory," in *Proc. 45th Annu. IEEE Int. Rel. Phys. Symp.*, Apr. 2007, pp. 167–170, doi: 10.1109/RELPHY.2007.369887.

[29] C.-P. Chen, H.-T. Lue, C.-C. Hsieh, K.-P. Chang, K.-Y. Hsieh, and C.-Y. Lu, "Study of fast initial charge loss and it's impact on the programmed states vt distribution of charge-trapping NAND flash," in *IEDM Tech. Dig.*, Dec. 2010, p. 5, doi: 10.1109/IEDM.2010.5703304.

[30] A. Maconi *et al.*, "Impact of lateral charge migration on the retention performance of planar and 3D SONOS devices," in *Proc. Eur. Solid-State Device Res. Conf. (ESSDERC)*, Sep. 2011, pp. 195–198, doi: 10.1109/ESSDERC.2011.6044201.

[31] K. Lee and H. Shin, "Investigation of retention characteristics for trap-assisted tunneling mechanism in sub 20-nm NAND flash memory," *IEEE Trans. Device Mater. Rel.*, vol. 17, no. 4, pp. 758–762, Dec. 2017, doi: 10.1109/TDMR.2017.2772046.